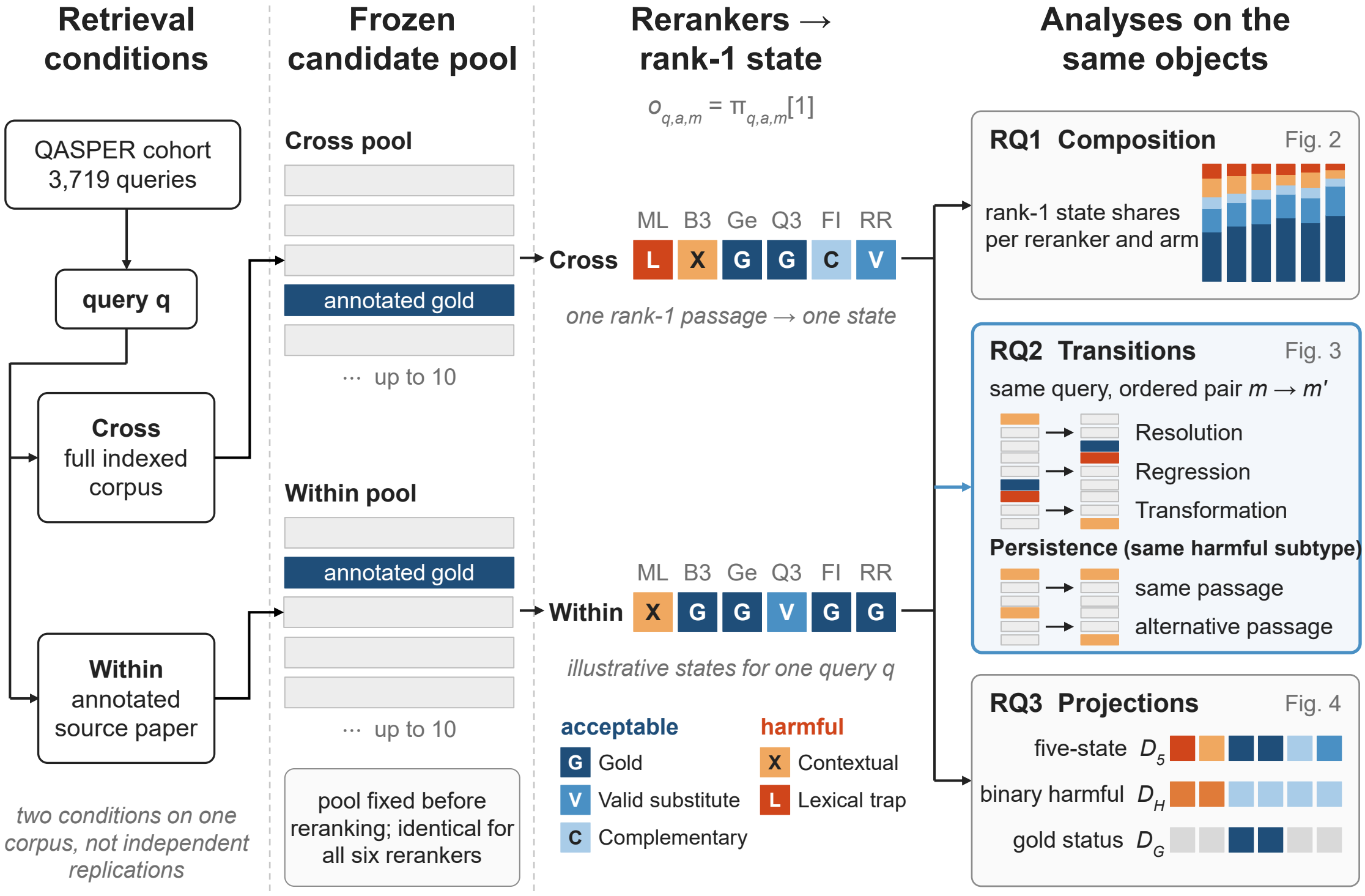

Retrieval conditions
Frozen candidate pool
Rerankers → rank-1 state
Analyses on the same objects
QASPER cohort 3,719 queries
query q
Cross full indexed corpus
Within annotated source paper
two conditions on one corpus, not independent replications
Cross pool
annotated gold
… up to 10
Within pool
annotated gold
… up to 10
pool fixed before reranking; identical for all six rerankers
$o_{q,a,m} = \pi_{q,a,m}[1]$
ML B3 Ge Q3 FI RR
Cross L X G G C V
one rank-1 passage → one state
Within X G G V G G
illustrative states for one query q
acceptable
G Gold
V Valid substitute
C Complementary
harmful
X Contextual
L Lexical trap
RQ1 Composition Fig. 2
rank-1 state shares per reranker and arm
RQ2 Transitions Fig. 3
same query, ordered pair m → m′
Resolution
Regression
Transformation
Persistence (same harmful subtype)
same passage
alternative passage
RQ3 Projections Fig. 4
five-state D_5
binary harmful D_H
gold status D_G